

**FINAL EXAMINATION
SEMESTER 1 2018/2019**

SUBJECT CODE : SCSR2033

SUBJECT : COMPUTER ORGANIZATION AND ARCHITECTURE

DURATION : 3 HOURS

DATE :

PLACE :

INSTRUCTIONS :

ANSWER ALL QUESTIONS FOR PART A AND PART B IN THE ANSWER BOOKLET.

(Please fill in your particulars)

Name						
Matric No.						
Years / Course						
Section						
Lecturer's name (Please tick.)	<table border="1"> <tr> <td>XXXX</td> <td>XXXX</td> </tr> <tr> <td>XXXX</td> <td>XXXX</td> </tr> </table>		XXXX	XXXX	XXXX	XXXX
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This questions paper consists of 10 printed pages, excluding this page.

Part A: Objectives (20 Marks)

1. Which of the following is **FALSE** about library procedure in assembly language?
 - A. Call a library procedure using the CALL instruction.
 - B. Some library procedures require input arguments.
 - C. The INCLUDE directive copies in the procedure prototypes.
 - D. Register EBC must be used in calling the library procedure.

2. All of the following are valid libraries procedure to be called **EXCEPT**:
 - A. Clrscr
 - B. DumpWatch
 - C. ReadString
 - D. Delay

3. What is the library procedure can be used to writes a null-terminated string to standard output?
 - A. WriteString
 - B. WriteInt
 - C. WriteBin
 - D. WriteChar

4. Which of the following is **FALSE** about the rule of logical instructions usage?
 - A. Operand 2 will be used as destination, while operand 1 as the source.
 - B. Operand 1 can be either register or memory.
 - C. Operand 2 can be immediate value only.
 - D. Memory-to-memory operations are not possible.

5. Which of the following is jump instruction that based on comparison of unsigned operands?
 - A. JGE
 - B. JNP
 - C. JAE
 - D. JNE

6. The _____ converts machine instructions of 0s and 1s into control signals.
- A. microprogram
 - B. control flags
 - C. micro sequencer
 - D. control unit
7. The problem of data dependency in instructions pipeline can be solved using _____.
- A. instruction forwarding.
 - B. operand forwarding.
 - C. delayed storing.
 - D. different data.
8. All of the following are input to the Control Unit **EXCEPT**
- A. status flags.
 - B. instruction register.
 - C. control signal within the processor.
 - D. control signal from control bus.
9. Which of the following is **NOT** an advantage of pipelining?
- A. Executing more than one task at a time.
 - B. Breaking the fetch-decode-execute cycle into tasks.
 - C. Finishing all current tasks before starting a new operation.
 - D. Allowing the processing of a new task even though the previous one has not ended.
10. The conflict of resources can occur in pipelining when _____.
- A. execution time is longer than the fetch time.
 - B. two pipeline segments need to access memory at the same time.
 - C. indirect cycle time is longer than the fetch cycle.
 - D. hardware interlock occurs.

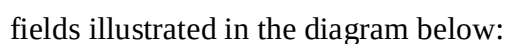
11. The following are character based I/O devices **EXCEPT**:
- A. modem.
 - B. mouse.
 - C. keyboard.
 - D. hard disk.
12. READ and WRITE are two common signals in processor communication with I/O module. These two signals are
- A. address signal.
 - B. data signal.
 - C. flag signal.
 - D. control signal.
13. The following statements are related to which data storage technology?
- Lifetime estimation of media is 100 years
 - Data is recorded in a single spiral track, starting from the centre of the disk and spanning outward
- A. Magnetic disk
 - B. Magnetic tape
 - C. Optical disk
 - D. RAID
14. The following statements are TRUE regarding the direct memory access **EXCEPT**
- A. CPU delegates the I/O processing to the I/O controller.
 - B. I/O processing done by files.
 - C. I/O controller has direct access to the main memory.
 - D. I/O controller sends an interrupt to the CPU to indicate success or failure.

15. The functions for I/O processors in channel I/O include:
- i. Negotiates protocols
 - ii. Issues device commands
 - iii. Translates storage coding to memory coding
 - iv. Delegates task for large files transfer to the host CPU
- A. All the above
B. i, ii and iii
C. ii, iii and iv
D. ii and iii
16. Which of these is a type of Input/Output (I/O) devices?
- A. Integer I/O devices
B. String I/O devices
C. Stream I/O devices
D. Block I/O devices
17. If access time for a cache is 15ns and a main memory access is 178 ns, meanwhile a hit rate of 97%. Compute the Effective Access Time (EAT):
- A. 172.66 ns
B. 14.55 ns
C. 19.89 ns
D. 5.34 ns
18. Miss penalties are usually much greater than hit times. Average Memory Access Time (AMAT) can be dropped by reducing
- A. Miss rate
B. Hit time
C. Memory access time
D. Cache access time

19. The following are the cache mapping scheme **EXCEPT**:
- A. Block direct mapping
 - B. Set associative mapping
 - C. Cache associative mapping
 - D. Fully associative mapping
20. Given address that access by the program is tag = 0000010, block = 100, word = 11.
What is the block size of cache?
- A. 3
 - B. 4
 - C. 8
 - D. 128

Part B (80 Marks)

A system uses a control memory of 2048 words of 32 bits each. The microinstruction has 3



- How many bits are there in the branch address field and the selection field? [4 marks]
- If there are 8 status bits in the system, how many bits are used to select an input for the multiplexers? [4 marks]
- What is the purpose of a control memory? [2 marks]

Assume a pipeline processor with 4 segments: instruction fetch (IF), operand fetch (OF), instruction execute (IE) and operand store (OS). Consider following sequence of instructions and answer the questions:

```

        AND AX, 011Fh
        SHL BX, 3000h
        JMP ABC
        ADD AX, CH
        SUB AX, 200
ABC:    MUL BX

```

- a. Draw the space time diagram for the above instructions. [4 marks]

- b. Solve the branching problem using NOP. [6 marks]

Question 3 (11 marks)

Given the data segment as below in assembly language. Write the simple coding of the following required standard output that based from the data segment. You might only need to write the important instruction(s) of the coding.

```
.data
    counter    DWORD    5
    value      DWORD    10h, 20h, 30h, 40h
    Answer     DWORD    0
```

- a. Add the `value` (10h + 20h), store the final result in `Answer`, and display the result in hexadecimal value. [5 marks]
- b. Generate and display 5 pseudorandom signed integers in the range 0 – 99. You must use the `counter` variable in your coding. Display each integer to a signed 32-bit integer to standard output in decimal format on a separate line. [6 marks]

Question 4 (9 marks)

Given the following pseudocodes with logical statements and compounded expressions. Convert them into assembly language by showing the `.data` and `.code` segments.

```
Marks = TotalMark = 60, Bonus = 5
; All are 16-bit unsigned integer

If (Marks > 50) and (Bonus = 5) then
    TotalMark = TotalMark + 2
Else TotalMark = TotalMark + 1
```


Question 5 (10 marks)

Table 1 shows the performance on two difference CPUs executing program C. For each CPU, answer the following questions:

- Calculate the Clock Cycles for each instruction type? [2 marks]
- Calculate the Instructions Per Clock Cycle (IPC) for each instruction type? [2 marks]
- Find the CPU clock cycles? [2 marks]
- Find the CPU execution time? [2 marks]
- Determine which CPU executes the instructions faster and by how much? [2 marks]

Table 1: The performance for CPU *Alien* and CPU *Smurf*

	CPU Alien		CPU Smurf	
Clock Rate	300 MHz		250 MHz	
Instruction Type	Instruction Counts	CPI	Instruction Counts	CPI
DIV	40	2.4	35	2.0
SUB	15	1.2	15	3.2
MOV	30	1.8	45	1.2
ADD	40	2.5	20	2.2

Question 6 (10 marks)

Input/Output (I/O) is define as a subsystem of components that moves coded data between external devices and a host system, consisting of a CPU and main memory. Answer the following questions.

- Computer systems employ any of four general I/O control methods. List down all those methods. [4 marks]

- b. Figure 1 illustrates the bus timing diagram that define bus operation in detail.

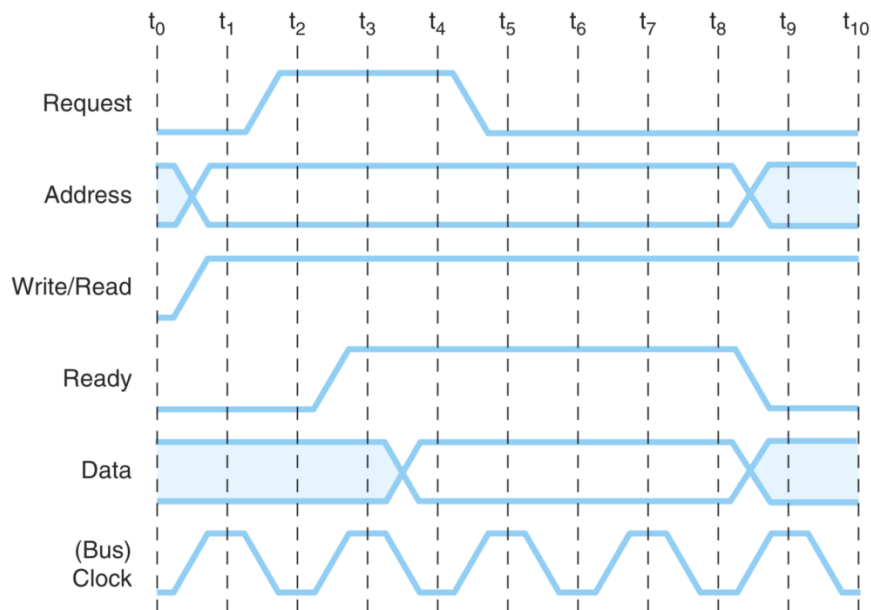


Figure 1: Bus timing diagram

Reconstruct and complete the Table 2 in the process of writing data to the disk drive.

[6 marks]

NOTE: Copy Table 2 into the answer booklet.

Table 2: Bus signals

Time	Bus Signal	Meaning
t_0		
t_1		
t_2		
t_3		
t_{4-7}		
t_8		

Question 7 (10 marks)

Given a memory address of 8 bits with 8 memory banks

- How many bits are needed to address the memory banks. [1 marks]
- Draw the LOI (Low Order Interleaving) address format. [2 marks]
- Calculate the memory bank capacity. [2 marks]
- Which module/bank that addresses 5Bh belongs to? [2 marks]
- Draw and label the sequence of address for the first and last module/bank using LOI. [3 marks]

Question 8 (10 marks)

Given that a *direct mapped* cache memory has 64KB data. The main memory capacity is 1MB.

- Find the number of bits for the main memory address. [1 marks]
- Find the number of bits for the cache memory. [1 marks]
- How many blocks are there in memory? [1 marks]
- Draw the main memory address format. [2 marks]
- Consider an 8 word direct mapped cache shown in Table 3. Show the contents of the cache as it responds to a series of request (decimal address) either it is hit or miss. [5 marks]

NOTE: Copy Table 3 into the answer booklet.

Table 3: A direct mapped cache memory

Decimal address	14	10	7	10	14	5	7	13	2
Binary address	1110	1010	0111	1010	1110	0101	0111	1101	0010
Hit or miss									
Line/slot address									

